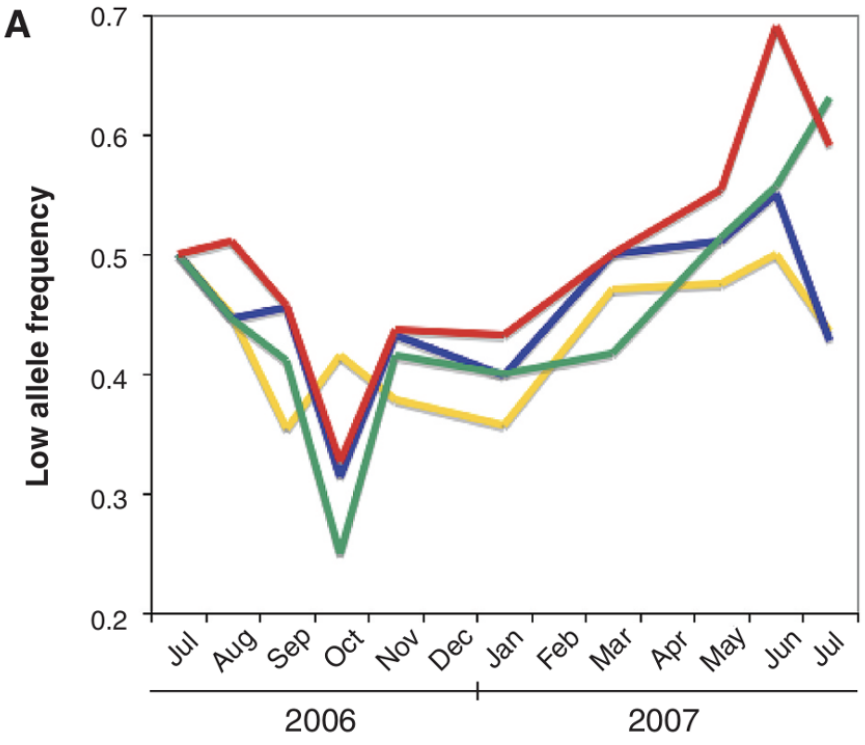
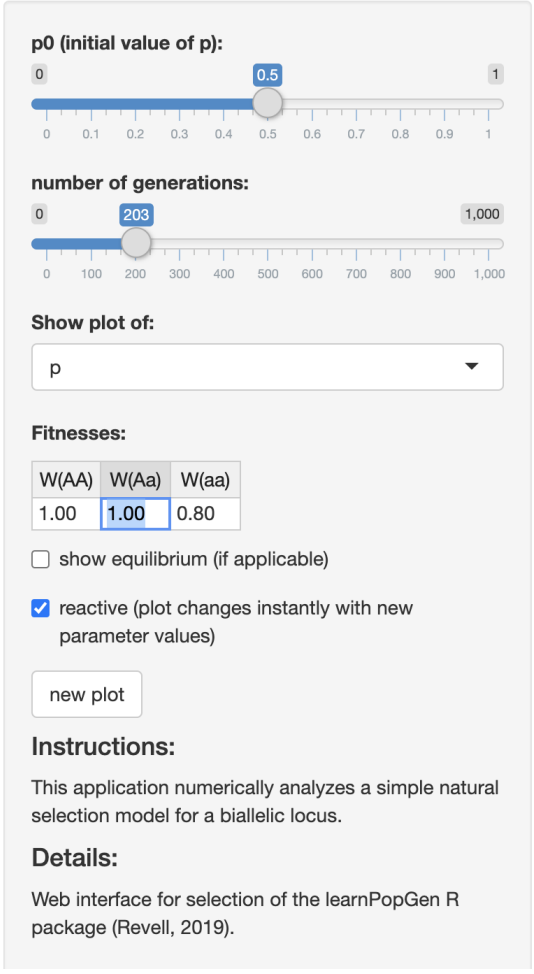
Natural selection in Threespine sticklebacks

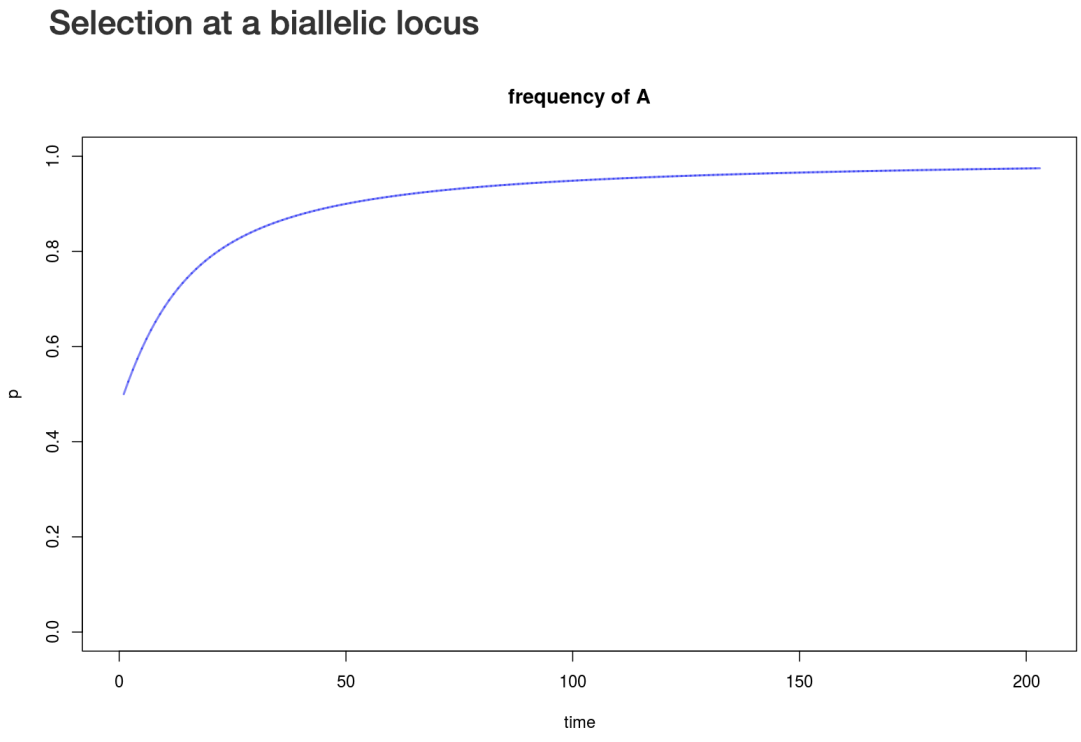




Selection at a biallelic locus

link: https://phytools.shinyapps.io/selection/





Selection Activity

1) Set fitnesses to WW(AA)=1.0, W(Aa)=1.0, and W(aa)=0.8. Set p0 to 0.01, and make sure you are showing a plot of p. What mode of inheritance is occuring at this locus? What is the value of the selection coefficient? How many generations does it take to reach p=0.6?

2) Change W(Aa) to 0.9. What mode of inheritance are you modeling now? Does the time it take to reach p=0.6 change?

3) Switch to showing a plot of q without changing any other settings. What can you conclude about the relationship between p and q through time?

4) Develop a question and address it using this app. Be prepared to report back to the class.

Nachman et al. Group Discussion

1) What is the mechanism of natural selection in this study?
2) What does it mean that MC1R is "perfectly associated" with coat color?
3) What does Table 2 tell us?
4) What does Figure 2 tell us?